



Handelshøyskolen BI

GRA 19703 Master Thesis

Final Thesis Master of Science ~~W~~ 100%

Predefinert informasjon

Startdato: 08-01-2024 09:00 CET
Sluttdato: 01-07-2024 12:00 CEST
Eksamensform: T
Termin: 202410
Vurderingsform: Norsk 6-trinns skala (A-F)
Flowkode: 202410||11436||IN00||W||T
External assessor: External assessor 1
Internal assessor: Internal assessor 1

Deltaker

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Informasjon fra deltaker

Tittel *:	The Venture Capital Puzzle
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Inneholder besvarelsen Nei
konfidensielt
materiale?:

Kan besvarelsen Ja
offentliggjøres?:

Gruppe

Gruppenavn: (Anonymisert)

Gruppenummer: 156

Andre medlemmer i gruppen:



The Venture Capital Puzzle

*Investigating The Potential Relationship Between Venture Capital Backing
And IPO Underpricing In European Markets (2000-2023)*

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Master thesis/GRA 19703

Master of Science in Business - Major in Finance

BI NORWEGIAN BUSINESS SCHOOL

This thesis was written as a part of the Master of Science in Business and major in finance at BI. Please note that neither the institution nor the examiners are responsible – through the approval of this thesis – for the theories and methods used, or results and conclusions drawn in this work.

Acknowledgements

We wish to express our gratitude to our supervisor Salvatore Miglietta at BI Norwegian Business School, whose expertise in the field of Corporate Finance significantly enriched our academic journey while shaping this thesis. His guidance and feedback have made the process of researching this topic both exciting and rewarding. We are also thankful to Einar Christopher Wellen for his assistance with our data analysis, and to Nenad Marovac, Founder and Managing Partner at DN Capital, for his insights into the venture capital industry, which have enriched our research.

BI Norwegian Business School

Oslo, July 2024

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Abstract

We study the relationship between venture capital backing and IPO underpricing in European markets from 2000 to 2023. Analyzing 6,637 IPOs with parametric and non-parametric tests, we find that VC-backed IPOs generally outperform non-VC-backed IPOs in first-day returns, both in the overall dataset and in a narrower time frame surrounding AIFMD's regulatory implementation in 2011. By dividing Europe into legal origins, we observe indications that higher-governance regions may experience lower underpricing post-law. The trend aligns with our hypothesis that increased market efficiency reveals the asymmetric information advantage that VCs can have. Despite these indications, the lack of statistical significance prevents us from conclusively supporting this hypothesis.

Abbreviations

AFDR Average first day return

AIF Alternative Investment Funds

AIFMD Alternative Investment Fund Managers Directive

DiD Difference-in-Differences

EVCA European Venture Capital Association

IPO Initial Public Offering

NVCA National Venture Capital Association

VC Venture Capital

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1 Introduction

1.1 Introduction & Motivation

The initial public offering (IPO) represents a significant milestone in a company's development, denoting its transition from private to public ownership. Our research will closely examine venture capital-backed IPOs, focusing on underpricing in the European market from 2000 to 2023. Underpricing refers to the phenomenon where shares are offered at a price lower than their market value (Chambers & Dimsion, 2009).

We will specifically analyze the change in the initial offer price to the final price recorded on the first trading day. The essence of this research lies in exploring the plausible causal relationship between venture capital backing and the underpricing of IPOs. This study is particularly relevant in the context of European markets, a region underrepresented in existing research. The noticeable absence of empirical evidence regarding VC-backed European IPOs, along with the recent increase in venture capital deals and the decline in the U.S. share of deals, underscores the relevance of this research area (Alex Warfel & Caroline Suttie, 2023). Most of the existing research has focused on the U.S. market due to its substantial volume of IPOs and intense venture capital activity. By shifting the focus to Europe, this study aims to fill a critical gap and broaden the understanding of global IPO dynamics.

1.2 Objectives & Research Question

It is crucial to determine the reasons behind the higher underpricing in venture capital-backed companies, as this represents a tangible cost to the venture capitalists due to the transfer of wealth to new shareholders, leaving the funders and VC firms in a position where there is money left on the table (Megginson & Weiss, 1991). Therefore, it becomes essential to investigate the underlying reasons that founders and venture capitalists carry this financial burden. The implications of this study should be multifaceted, providing

insights for companies considering public offerings, investors navigating IPO markets, and policymakers aiming to regulate and stabilize financial markets.

1.3 Causality & the larger picture

To isolate the effects of venture capital backing on IPO underpricing, our study will employ statistical analysis to explore this potential connection. We aim to identify not only correlations but also causality, utilizing a difference-in-difference analysis to examine any potential causal relationships. By comparing IPO underpricing between VC-backed and non-VC-backed companies in response to a regulatory shock in European markets that mainly affect venture capital, we seek to isolate the potential causal effects of VC backing on first-day stock performance. Examining the price sensitivity to regulatory shifts, focusing on the Alternative Investment Fund Managers Directive (AIFMD) implementations following the financial crisis of 2008.

Additionally, utilizing the findings from Law and Finance by Rafael La Porta (Porta et al., 1998), we aim to assess how this impact may vary with the level of legal origin within European countries. Ultimately, this thesis will contribute to the broader understanding of IPO mechanics, particularly the influence of venture capital. If a link between VC backing and IPO underpricing is established, it could have significant implications for companies considering going public and for investors looking to capitalize on potential positive first-day returns.

2 Literature review and theory

2.1 Literature review

In this section, we review existing research papers and studies that provide essential insights into the issues related to our research question. Most of the data on initial public offerings with venture capital are centered around the American market, due to its substantial venture capital activities (Alex Warfel & Caroline Suttie, 2023). Furthermore, much of the documentation found is not up-to-date, and many of these studies were conducted before the year 2000. Leaving out significant market changes and crises, additionally to the incorporation of new laws to obtain more efficient markets.

Significance of Venture Capital

Venture capitalists play a vital role in guiding high-potential companies, often acquiring significant equity stakes that grant them considerable influence over management decisions. Christopher B. Barry, characterize these investors as providers of high-risk capital to promising startups (Barry et al., 1990). Their overarching goal is to maximize financial returns by exiting investments through strategic sales or initial public offerings. This active ownership model is anticipated to yield superior returns compared to direct investments in companies (Metrick, 2006).

Prior to making an investment, venture capitalists and startups undergo a substantial due diligence process. This rigorous evaluation provides VCs with a comprehensive understanding of the startups, enabling them to make informed investment decisions based on non-public information. While this process may appear to suggest market inefficiency due to VCs' reliance on exclusive insights, previous research done by Wolfgang Bessler and Martin Seim show that other institutional investors in high-risk startups follow similar due diligence practices (Bessler & Seim, 2012). Therefore, there is minimal risk of endogeneity or selection bias in VC startup selection.

Leading on Bessler and Martin’s findings, Joel A.C. Baum’s research explores whether VCs act as effective scouts or coaches, debating if VC’s abnormal mispricing in IPOs is due to their ability of identifying winners or by building them (Baum & Silverman, 2004). Where VCs leveraging their networks and specialized knowledge to support startups more effectively. The findings suggest that VC expertise is indeed beneficial, but only to a certain extent. VCs’ financing decisions appear to be influenced by cognitive biases that lead to an overemphasis on the startups’ human capital. However, due to a small data sample and low statistical significance, we exercise caution in placing excessive weight on these findings. Nevertheless, this studies showcases that a startup is likely to select VC funding based on their sector knowledge and market connections within there investment mandate (Baum & Silverman, 2004).

Research conducted by Ewens and Malenko emphasizes the substantial influence of venture capital investments on IPO first-day returns. Using a dataset of 7,780 VC-backed startups from 2002 to 2017, their study analyzes board composition dynamics through regression analysis and difference-in-differences design. They find that VCs enhance governance by appointing independent directors who mediate and advise, making startups more attractive to investors and boosting first-day trading prices, aligns with Joel A.C. Baum’s findings (Ewens & Malenko, 2020).

The research further highlights that VCs provide strategic and financial expertise in IPO preparation. This expertise, combined with the credibility of reputable VCs, attracts significant investor interest and enhances first-day returns. Additionally, VCs’ focus on aggressive growth strategies aligns with investor preferences, creating a buzz around the IPO (Ewens & Malenko, 2020).

Ewens and Malenko also find that VCs may strategically underprice IPOs to ensure successful market debuts, aligning with Lee and Wahal’s grandstanding theory. This underpricing boosts demand, leading to higher first-day returns and serving as a strong market signal of startup quality,

further amplifying investor interest and confidence (Ewens & Malenko, 2020) (Lee & Wahal, 2004). Thus, strategic expertise, and intentional underpricing contribute to higher first-day returns in VC-backed IPOs.

Historical Underpricing

Underpricing, defined by Berk and DeMarzo, represents a significant source of indirect costs to issuing companies, evidenced by the 17% average first-day return in the US market from 1960 to 2015 (Berk & DeMarzo, 2017). This substantial underpricing implies that companies fail to fully capitalize on market valuations, with Loughran and Ritter quantifying the loss at an average of \$9.1 million for the US market, twice the average fees paid to investment bankers (Loughran & Ritter, 2002).

To further highlight potential winners from underpricing in an IPO, we need to mention the role of underwriters. Underwriters, who purchase securities from the issuing company and sell them to investors, expose themselves to risk through firm commitments. If they overestimate market interest, they risk losses, potentially selling shares at lower prices than purchased (Banton, 2023). This creates a principal-agent problem where underwriters underprice to minimize their exposure, aligning with Loughran and Ritter's findings that only 9% of US IPOs between 1990 and 1998 experienced a decline in share price on the first trading day, indicating underpricing tendencies (Loughran & Ritter, 2002).

Inconsistent observations

Meggison and Weiss delves into the impactful role of venture capitalists in the IPO process, highlighting their significant contribution not only in reducing the overall costs of public offerings, but also in maximizing the net proceeds for issuing firms (Meggison & Weiss, 1991). A key finding of the study is the tangible impact of VC involvement on IPO underpricing. The research shows that VC-backing effectively lowers both the average and median levels of IPO underpricing. This is particularly important as it

points to a more efficient pricing strategy in VC-backed IPOs, compared to their non-VC-backed counterparts.

The study also sheds light on the initial returns from these IPOs, revealing a lower mean initial return for VC-backed IPOs of 7.1% compared to 11.9% for those without VC backing (Megginson & Weiss, 1991). This difference is critical as it indicates that VC-backed IPOs experience less underpricing and, consequently, lower underwriter compensation. This finding remains robust even after accounting for variables such as the size of the offering, the quality of the underwriter, and the age of the firm. The study uses IPO data spanning from 1983 to 1987, thus covering a relatively brief period while also focusing exclusively on the US market (Megginson & Weiss, 1991). However, the dataset is both outdated and limited to a short time span, constraining its capacity to illustrate longer-term trends and diminishing the robustness of its conclusions.

Similar to the findings of Megginson & Weiss, the study by Barry looks into the influence of venture capital quality on IPO underpricing. Their research suggests that the capital markets acknowledge the value of high-quality VC monitoring services, as evidenced by the reduced underpricing in IPOs associated with top-tier venture capitalists (Barry et al., 1990).

They look into whether the capital markets recognize and reward the monitoring role played by venture capitalists. Their findings indicate that IPO's linked with higher-quality VC's display less underpricing, underscoring the significant impact these VC's have in the public market perception. The research underscores the crucial role of venture capitalists as major stakeholders in shaping and guiding new public companies. Their presence as large stockholders is not only a financial boost, but also a governance and monitoring asset, which is valued by the capital markets and reflected in more favorable outcomes in the IPO pricings (Barry et al., 1990).

2.2 Recent data trends

In our search for newer and more comprehensive research papers, we came across the study by Lee and Wahal. The study is more comprehensive, with a greater dataset of 6,413 IPO's from 1980 to 2000 in the US market. Lee and Wahal's finds a notable pattern observed in the significant clustering of VC-backed IPO's across specific industries, predominantly in technology-intensive sectors like software (Lee & Wahal, 2004).

The findings diverge from previous research regarding the impact of VC backing on IPO underpricing. Contrary to earlier studies that suggested VC backing reduces underpricing, the data in this study indicate that VC-backed IPO's actually experience greater underpricing.

The sample size is considerably larger and covers a longer time period compared to the other studies mentioned. These findings are not only statistically significant but also economically substantial, especially considering that the average underpricing of VC-backed IPOs over this period is 26.8%. This aligns with new findings from Ritter in 2023, with 9127 observations from 1980 to 2022 displaying a 27.2% average underpricing for VC-Backed IPOs (Ritter et al., 2023).

Lee and Waha's findings also suggest that younger venture capital firms tend to favor underpricing in IPOs as a strategy to boost their growth and establish their reputation in the market called grandstanding. This approach contrasts with the strategies of more established VC firms and show how there are other perspective on the factors influencing IPO pricing (Lee & Wahal, 2004).

A noteworthy study on the European IPO market by Wolfgang Bessler and Martin Seim, covering the period from 1996 to 2010, primarily focuses on the long-term returns of VC-backed IPOs. The study demonstrate a systematic underpricing trend over an extended period, with a noticeable underpricing tendency measured at 8.39% compared to 5.12% of non-VC backed (Bessler & Seim, 2012).

The substantial differences in the previously established results, underscores the need to consider a broader array of factors when assessing the impact of VC-backing in IPO's. The existing literature, consist of mainly US data as well as older data sets that fails to incorporate the increasing influence of venture capital in more recent years.

Consequently, it appears to be a noteworthy gap in current research, particularly concerning the rise of venture capital in recent times and it's specific effects within the European market. A summary of the former research and their respective findings is provided in Table 2.1¹.

Author	Years Analyzed	Non-VC	VC
		Backed (%)	Backed (%)
Ritter (2023)	1980–2022	13.5 %	27.2 %
Bessler & Seim (2012)	1996–2010	5.12%	8.39%
Lee & Wahal (2004)	1980–2000	19.36%	26.82%
Franzke, S. A. (2003)	1997–2002	48.38%	52.44%
Meggison & Weiss (1991)	1983–1987	11.9 %	7.1 %
Berry et al. (1990)	1978–1987	7.47%	8.43%

Table 2.1: Comparison of non-VC-Backed vs. VC-Backed Returns from the Literature

¹Bessler & Seim (2012) focus on venture capital-backed returns in Europe, with particular emphasis on major markets such as the UK, Germany, France, and Switzerland

Change In Global Deal Count

Table 2.2 highlights significant shifts in global deal counts over recent years. In 2005, the U.S. accounted for 71% of global deals, but this share has dropped dramatically to 37% by 2022. Additionally, the global deal count has increased from 4,325 in 2005 to 44,491 in 2022. Due to this change in global deals, the European market has become more relevant. This study will explore the potential variations in geographic locations and the implications of this recent data in the European context (Alex Warfel & Caroline Suttie, 2023).

Year	Global Deal count	US as % of Global Venture Activity
2005	4,325	71%
2006	5,248	65%
2007	6,988	64%
2008	7,626	64%
2009	7,394	62%
2010	9,654	58%
2011	12,222	57%
2012	14,601	55%
2013	18,771	53%
2014	23,674	47%
2015	29,474	40%
2016	30,053	36%
2017	32,188	36%
2018	35,952	35%
2019	37,253	36%
2020	38,208	35%
2021	51,601	36%
2022	44,491	37%

Table 2.2: Global Venture Activity In Recent Years

3 Data Collection & Cleaning

In order to conduct a comprehensive examination of our hypotheses, it is essential to collect data from a variety of databases. Our data collection is primarily concentrated on acquiring information regarding companies within the European continent that transitioned to public ownership from 2000 to 2023. Understanding the details of public market dynamics within this specific geographical and time-based context is critical, as this dataset will serve as a cornerstone for our analysis.

Our research approach involves collecting data that includes information on investor types, sectors, and industries. Here, we delve into the various industries these companies are part of, to better understand the market segments driving public ownership transitions. This analysis enables us to more effectively identify the factors influencing these transitions.

The data analysis will begin with the cleaning of data to ensure consistency across different sources. We will perform statistical tests to verify the significance of the possible abnormal returns observed. These tests will include parametric and non-parametric tests to account for the potential non-normal distribution of returns (Brown & Warner, 1985). Additionally, we conducted an interview with Nenad Marovac, Founder and Managing Partner of DN Capital and former Chairman of Invest Europe (EVCA)², to gain his perspective on country-specific investment strategies and industry insights.

Furthermore, we integrate the country-specific classifications and numeric ratings by Rafael La Porta on legal origin and investor protection scores within Europe. The Law and Finance article findings consist of a complete dataset of 49 countries. From this dataset, we filtered out 17 European countries and classified them by legal origin: Common Law, French Civil Law, German Civil Law, and Scandinavian, following the classification by Rafael Porta (Porta et al., 1998).

²EVCA - European Venture Capital Association

This classification allows us to compare countries within similar legal frameworks, providing insights into the impact of legal protections on market efficiency.

Common Law Countries: Strongest investor protections (e.g., UK, Ireland).

French Civil Law Countries: Weakest investor protections (e.g., Netherlands, France).

German and Scandinavian Civil Law Countries: Intermediate protections (e.g., Switzerland, Austria, Denmark, Finland).

Outliers, such as the Netherlands and Switzerland, which exhibit high enforcement quality despite their Civil Law origins, underscore the necessity of considering individual country characteristics. To address the absence of a general score across the four categories, we calculated the average weighted scores for each country. This reclassification aligns outliers more closely with the average scores of their respective legal origin groups, thereby enhancing the robustness of our analysis by standardizing comparisons across all factors between different legal origins (Appendix table A.6).

Data Gathering

This chapter is dedicated to outlining the sources from which this data was obtained and will provide a detailed elaboration on the methodologies employed in the processes of data gathering and cleaning.

To secure a significant dataset, it's essential to acquire comprehensive information about IPOs conducted within Europe during the specified time frame. For this purpose, the Bloomberg terminal stands out as an invaluable resource, having an exclusive database that includes data relating to nearly all IPOs conducted, making it an ideal source for our research. The Bloomberg terminal, well-known for its extensive financial data and analytics capabilities, offers real-time and historical information about financial markets and securities worldwide, which is crucial for our analysis.

From the Bloomberg database, we obtained a dataset which includes both venture capital-backed and non-venture capital-backed IPOs. This dataset consists of 6,637 observations and includes critical information such as sector and industry data, post-IPO returns during different time periods, and country-specific information, among other details (Appendix table A.1).

Further, we addressed possible gaps in the Bloomberg dataset by integrating information from the PitchBook database. PitchBook is well-known for its extensive coverage of global venture capital, private equity, and mergers & acquisitions, making it essential to our data gathering process. This was achieved by connecting to the PitchBook database via FileZilla and downloading a comprehensive dataset. We then used DBeaver, a platform used to manage data with SQL, to extract the necessary information. Specifically, we queried the database tables 'Company' and 'CompanyInvestorRelation' (Appendix table A.2).

It is important to note that for the purpose of our study, 'Europe' is defined according to the countries located in the European continent with the exclusion of Russia. This exclusion is due to unique country-specific effects and the presence of statistical outliers within the Russian market, such as instances of unrealistically high first-day returns, which could distort the overall dataset.

Our initial data review involved converting raw data from CSV format into a structured format using Excel Query, ensuring that rows and columns were correctly aligned for seamless integration into our analysis framework. We merged the datasets from Bloomberg and PitchBook in Excel Queries before importing them into Python.

This merging process used the company ticker as a unique identifier to mitigate issues arising from multiple tickers or ticker changes over time. In instances of overlap, data from Bloomberg were prioritized due to its comprehensive nature, leading to the exclusion of duplicated entries from PitchBook. Subsequently, we imported the refined data into Python using the Pandas library, streamlining the subsequent data cleaning steps.

The combined dataset was merged with the legal origins cited in Law and Finance, matching the countries and sourcing them into their correct legal origins. Due to the inclusion of ALPHA-2 ISO country codes in both datasets, it simplified the process of associating the appropriate national legal systems scores with each IPO ³. This approach resulted in a robust dataset, providing a solid foundation within the European context (Porta et al., 1998).

Legal Origin	German Civil Law	Common Law	French Civil Law	Scandinavian Law
Country	Germany	United Kingdom	France	Sweden
	Austria	Ireland	Italy	Denmark
	Switzerland		Spain	Norway

Table 3.1: A Sample of Countries and Their Legal Origins

Data Cleaning

The cleaning of data constitutes a critical step in our research, given the potential for inconsistencies and errors inherent in large datasets. Our initial action involved the exclusion of rows where 'Offer To 1st Close' was not available (NaN), thereby eliminating IPOs lacking viable data. This process ensures that our analysis is based on complete records. Additionally, we refined our dataset by removing entries associated with countries not falling within our predefined European scope.

An essential aspect of our dataset was the 'Offer Type' category, which defined the nature of each public offering, including differences between private equity-backed, venture capital-backed, and other types of backing. By filtering this data to retain only those IPOs categorized as VC-backed, we generated a new dataset focused specifically on venture capital influence.

³Alpha-2 codes are two-letter country codes defined in ISO 3166-1 by the International Organization for Standardization.

To identify and address outliers within our dataset, we employed the Python library `matplotlib.pyplot` and `seaborn` for data visualization. By creating heatmaps, bar charts, and histograms to distribute IPO occurrences by country, industry and first day return, we were able to visually detect anomalous data points⁴.

Subsequently, we conducted a manual review of the top and bottom 50 performing entries based on first day returns. This meticulous examination led to the removal of certain records where discrepancies or erroneous data were found, particularly those that exhibited inaccurate pre-IPO price. By removing these outliers, we enhanced the dataset's accuracy and relevance, aligning it more closely with real-world scenarios and ensuring that our analysis reflects genuine market conditions.

Selection Bias

While the process of data cleaning ensures data accuracy and relevance, it inevitably introduces the potential for selection bias. Selection bias occurs when certain groups within a dataset are more likely to be included or excluded, leading to non-representative samples that can skew research findings (Linton & Shankar, 2023).

In our study, data were sourced from trustworthy databases such as Bloomberg and PitchBook. Despite the robust nature of these sources, the act of excluding certain companies during the cleaning process, whether due to missing data, country exclusions, or the removal of statistical outliers, can result in a dataset that does not fully represent the broader population of companies undergoing IPOs within Europe. Liu and Ritter highlight that such exclusions and the selection criteria used can significantly impact the representativeness and findings of IPO studies (Liu & Ritter, 2011).

⁴The filtered European dataset includes the following countries: Austria, Azerbaijan, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, North Macedonia, Norway, Poland, Portugal, Romania, Serbia, Slovenia, Spain, Sweden, Switzerland, Turkey, United Kingdom.

The most significant concern arises from the exclusion of IPOs lacking 'Offer To 1st Close' data or those with large abnormal first-day returns. Such exclusions, while necessary to maintain data integrity, may disproportionately affect smaller companies or those in less prominent sectors, which are more likely to have incomplete or irregular data (Gompers & Lerner, 2003).

Consequently, our dataset may be biased towards larger, more established companies with comprehensive data records. This bias could impact the generalizability of our findings, as the financial behaviors and post-IPO performances of these larger entities might differ systematically from those of smaller firms (Loughran & Ritter, 2004).

While the data cleaning is essential for ensuring the reliability of our dataset, it is important to acknowledge the potential for selection bias. Despite this, such filtering processes are essential to perform to maintain the integrity of our data and ensure its relevance.

4 Hypothesis & Methodology

4.1 Hypotheses

Venture Capital's Influence on IPO's

Efficient market theory suggests that securities should be priced accurately, reflecting true value immediately post- and pre-IPO (Lowry & Schwert, 2004). Yet, empirical evidence indicates that IPO's often experience systematic mispricing. Previous research into the impact of venture capital backing on IPO mispricing presents mixed findings, influenced by geographic and methodological variables. Our hypothesis, based on the literature review, suggests that in Europe, IPO's with venture capital backing might tend to achieve greater first-day average returns compared to those without VC backing.

Hypothesis 1: The first day mean return for European venture-capital-backed IPOs is greater than zero.

Hypothesis 2: The first day mean return for European venture-capital-backed IPOs is greater than that for non-venture-capital-backed IPOs.

Causality of VC backing on IPO first-day returns

Findings from the literature review indicates that venture capital backing may not only correlate with, but could actually cause an increase in the first day mean return of IPOs. There could be a potential causal effect rooted in the added value that VCs bring to the table, such as enhanced credibility, better access to key resources, and strategic guidance, as noted by Megginson & Weiss and Gompers (Gompers, 1996) (Megginson & Weiss, 1991). In the context of European markets, where regulatory frameworks and investor profiles can vary significantly, the impact of VC backing on IPO performance might have differed in recent years due to the increased number of global deals (Alex Warfel & Caroline Suttie, 2023).

A hypothesis that tests this is crucial for analysing the mechanisms driving underpricing tendencies in IPO markets, as it offers insights into how regulatory interventions shape market behaviors and outcomes. By demonstrating a strong effect of the law on mispricing levels in venture-backed IPOs relative to a control group, this hypothesis could indicate correlation and evidence of a causal effect in systematic underpricing.

Hypothesis 3: In European markets, IPOs with VC backing demonstrate less underpricing than the control group after the implementation of law 32011L0061 stated by the AIFMD in 2011.

To further highlight the effects on the control group, we aim to examine country-specific outcomes in Hypothesis 4. This hypothesis shifts the focus to the role of demographic factors in shaping the post-law underpricing landscape across European countries. Building upon previous research indicating the significance of legal financial factors. This study hypothesizes that countries with higher governance will demonstrate lower levels of underpricing following the introduction of the new law compared to countries with lower governance scores.

Hypothesis 4: After the implementation of law 32011L0061, regions with higher governance will experience significantly lower levels of underpricing compared to regions with lower governance.

Hypotheses 4 offer a comprehensive framework for investigating the interplay between regulatory changes, demographic, and financial factors in the European IPO market. This framework aims to provide valuable insights into the drivers of underpricing and the effectiveness of regulatory interventions in enhancing market efficiency. Ultimately, these hypotheses will deliver empirical indication on the causal relationships between venture capital's influence and IPO mispricing. Together, they will offer valuable evidence for relationships towards VC's effect towards IPO underpricing.

4.2 Methodology

This chapter outlines the methodology adopted to examine the influence of venture capital backing on the initial pricing and subsequent performance of IPOs in Europe. It details the research approach, study design, and the empirical methodology, along with the justification for utilizing the legal regulation implemented by AIFMD in 2011 as a market shock.

4.2.1 Research strategy

In connecting the theoretical hypotheses with empirical data, this study adopts a deductive approach as described by Mark Saunders. This approach begins with an established theoretical framework to formulate hypotheses. Subsequently, data is systematically collected and analyzed to test these hypotheses. The deductive approach allows for a structured investigation where the research hypotheses, derived from existing theories on IPO performance and VC influence, can be empirically validated (Saunders et al., 2016). Should the findings be inconclusive, the study may adopt an inductive stance to interpret the results within the context of existing theories, and to explore the potential for new theoretical developments.

4.2.2 Research design

Given the objective of our study to quantify the effect of VC backing on IPO underpricing and performance in Europe, a quantitative research method is most appropriate (Johannessen et al., 2021). Specifically, an event study methodology will be employed to evaluate the instant impact of VC backing on IPO share prices.

This method measures possible abnormal returns attributed to the IPO event to isolate the effect of VC backing from broader market movements. Event studies, based on the efficient market hypothesis, assume that stock prices reflect all available information (Fama, 1970), making them ideal for assessing market reactions to corporate events. This approach provides

insights into how investors value the presence of venture capital in the IPO process and the efficiency of market pricing before and after the IPO phase.

4.2.3 Event window & definition

In our study, the IPO listing day serves as the event day when the stock price is disclosed, and the influence of venture capital on stock prices becomes observable (Beatty & Ritter, 1986). The event day represents the final anticipatory activities by VCs and is when the market responds to the newly available information about the IPO (Fishe et al., 2019).

However, due to the nature of IPOs, our event window can only feasibly extend backward to one day prior to the listing, with the signing price as the first observable valuation. We will employ a single-day event window, which includes the day before the listing (-1) and the public listing day itself (1), to capture the immediate market reaction. Post-listing, we will observe a short-term window ($-1, 1$) days to measure the market response to the IPO and VC involvement. This approach aligns with the work of Bradley, Jordan, and Ritter, who discuss the informational efficiency of IPO markets and the rapid integration of new information into stock prices (Bradley et al., 2004).

4.3 Methodology for testing abnormal returns

The immediate window is expected to show a significant reaction if the market perceives VC backing as a positive signal, reflecting higher confidence in the IPO's prospects. The post-announcement window will provide insights into the sustained effect of VC backing as the market continues to adjust to the new information.

Hypothesis 1: *The first day mean return for European venture capital backed IPOs is greater than zero.*

To ensure the robustness of our findings, we will utilize both parametric and non-parametric statistical tests. Initially, we utilize the Shapiro-Wilk test to assess the normality of the dataset (M, Tsagrisa et al., 2020).

Given the importance of validating the normality assumptions, we apply a one-sample t-test for parametric analysis, contingent on the normality of the data. Simultaneously, the non-parametric Wilcoxon signed-rank test is used to address any deviations from normality.

The Central Limit Theorem (CLT) underpins our parametric analysis, positing that with sufficiently large sample sizes ($n > 30$), the sampling distribution of the mean approximates normality regardless of the original population distribution (Kwak & Kim, 2017). This principle justifies the use of the t-test given our dataset's sample sizes, which exceed this threshold.

In this study, the average first day return is calculated for each venture capital-backed firm using the formula underneath. Here, the 'First day closing price' represents the closing price on the first day of trading, and the 'Offer Price' denotes the initial offer price for the firm.

$$\text{Average First-Day Return} = \frac{1}{n} \sum_{i=1}^n \left(\frac{\text{First day closing price}_i}{\text{Offer Price}_i} - 1 \right)$$

Hypothesis 2: *The first day mean return for European venture capital-backed IPOs is greater than that for non-venture-capital-backed IPOs.*

The hypothesis can be formulated as:

$$H_0 : \bar{R}_{VC} < \bar{R}_{NVC}$$

$$H_1 : \bar{R}_{VC} > \bar{R}_{NVC}$$

To test the second hypothesis, we started by using Levene's test, a statistical process used to assess the equality of variances for the two groups. Then we tested the dataset for statistical significance in the first-day mean returns between venture capital-backed IPOs and non-venture-capital-backed IPOs, using both Welch's t-test for unequal variances and the Mann-Whitney U test. We let $\bar{R}_{VC}$ represent the mean first day return for venture capital-backed IPOs, and $\bar{R}_{NVC}$ represent the mean first day return for non-venture-capital-backed IPOs.

Hypothesis 3: Utilizing an independent two-sample t-test and a difference-in-differences regression framework, we aimed to assess the impact of the Alternative Investment Fund Managers Directive law 32011L0061 of 2011 on mispricing levels in European IPO markets (“Directive 2011/61/EU”, 2011). Given the suggesting presence of heteroscedasticity in the data, we used robust standard errors to account for this in the DID. Additionally, we will use T-tests and Mann–Whitney U tests to validate the results.

Establishing causality is challenging in research methodology, particularly because we cannot randomly allocate startups to receive VC backing. Controlled studies, where samples are split into comparable groups receiving different treatments, are an ideal method for proving causality (ABOS, 2023). Yet, in this context, random allocation is not feasible. Therefore, we will implement a quasi-experimental approach, designating IPOs without VC backing as the control group, while VC-backed IPOs will serve as the treatment group.

To capture potential market trends surrounding the legal shock in 2011, we use a time frame of three years before and three years after the event. This approach allows us to observe both short-term and long-term developments, enabling us to identify possible trends and potential normalization in the market. Given that the average timeframe for IPOs is approximately six to nine months, this observation period provides a comprehensive view of the market dynamics influenced by the legal shock (PitchBook, 2023).

The quasi-experimental method isolates the treatment effect of AIFMD’s law on venture-backed IPOs relative to non-venture-capital-backed IPOs⁵. We will test the means of two independent groups, pre-law and post-law periods, to determine if they are statistically different. While the DiD approach allows us to control for time-invariant unobservable factors and potential biases, it should be noted that concerns of endogeneity may still arise from factors such as non-random assignment and effect of venture capital backing (Bartolucci et al., 2015).

⁵This dataset includes only countries affected by the AIFMD regulation.

The DiD model is specified as follows:

$$\text{Mispricing} = \beta_0 + \beta_1 \text{PostLaw} + \beta_2 \text{VCBacked} + \beta_3 (\text{PostLaw} \times \text{VCBacked}) + \epsilon \quad (4.1)$$

Where:

- Mispricing represents the IPO mispricing.
- β_0 represents the intercept or mean return of Non-VC backed IPOs.
- PostLaw is a dummy variable indicating the period after the implementation of the regulatory changes presented by AIFMD.
- VCBacked is a dummy variable indicating whether the IPO is venture capital-backed.
- $\text{PostLaw} \times \text{VCBacked}$ is the interaction term to measure the differential impact on VC-backed IPOs post-law.
- ϵ is the error term, capturing unobserved factors affecting IPO mispricing.

4.4 Corporate governance in Europe

For the methodology of **Hypothesis 4**, we categorized European countries into four legal origins based on their shareholder, creditor, accounting and enforcement scores. We then applied the same regression model as in H3 (4.1) and conducted a DiD analysis for each legal region (Appendix Table A.6).

By examining these effects across different legal origins, we can understand how legal contexts moderate the impact of regulatory changes. This highlights whether VC-backed IPOs had higher first-day returns compared to non-VC-backed IPOs before the treatment and if there were any significant changes afterward.

By integrating La Porta's findings, we support our hypotheses that the legal and socio-economic frameworks of countries significantly influence their market dynamics and willingness to adopt new regulations. Specifically,

countries with stronger legal protections for investors, are more likely to demonstrate reduced mispricing post-regulation (Porta et al., 1998).

4.5 AIFMD as a Regulatory Shock

To contextualize our findings, it is essential to understand the background and historical events that led to the regulatory changes used to demonstrate causality in our analysis.

The 2008 financial crisis exposed significant weaknesses in the global financial system, underscoring the urgent need for enhanced market efficiency and transparency. The crisis, triggered by the collapse of major financial institutions and the subsequent economic downturn, highlighted the necessity for stricter regulations for alternative fund managers and other financial institutions. This regulatory overhaul aimed to mitigate systemic risks and prevent future financial instability, thereby fostering a more transparent and stable market environment, which is crucial for reducing mispricing in IPOs.

Time for regulatory changes leading to AIFMD

In response to the 2008 financial crisis, the Alternative Investment Fund Managers Directive was proposed to the European Union through the regulation change document 52009PC0207 in 2009. This proposal aimed to raise availability to the vulnerabilities revealed by the crisis by decreasing the likelihood of future financial instability. AIFMD was formally presented and adopted 08/06/2011 with the validation of document 32011L0061 in EU, establishing a comprehensive regulatory framework for alternative investment fund managers.

The directive's goal was to enhance transparency, improve investor protection, and bolster market integrity. Although the formal implementation across EU countries was scheduled for July 2013, the directive's effects on the market were immediate, influencing behaviors and practices from the moment it was approved in 2011. Despite countries being allowed to wait until 2013 to announce it in their own jurisdictions ("Directive 2011/61/EU", 2011).

Document 32011L0061, introduced in 2011, was a major change from the initial suggested regulation presented in 2009. This difference was quite surprising due to its significant impact and the lack of transparency leading up to the law. The specifics of these changes are detailed in section 4.6 below.

4.6 AIFMDs effect on VCs

AIFMD plays a crucial role in regulating the activities of Alternative Investment Fund Managers, particularly those involved in venture capital investments. Specific provisions within the AIFMD are directly relevant to the involvement of venture capitalists in IPOs, ensuring greater transparency, accountability and financial stability. These provisions are instrumental in reducing mispricing in VC-backed IPOs by fostering a more stable and transparent market environment.

Transparency and Reporting Requirements

Article 26 outlines the conditions under which AIFMs managing AIFs that acquire control over non-listed companies must operate. It emphasizes transparency and detailed reporting requirements, which is critical for maintaining market integrity. According to Article 26(1), AIFMs must disclose acquisitions to the authorities and relevant stakeholders, ensuring that all significant transactions are transparent and properly documented (“Directive 2011/61/EU”, 2011). This is leading to a reduction in information asymmetry, supporting our hypothesis that regulatory changes lead to less mispricing.

Detailed Information and Disclosure

Building on the transparency framework, Article 27 mandates that detailed information about the acquisition, including the identity of shareholders and the means of financing, must be provided to the authorities and employees. This level of detail helps in mitigating risks associated with hidden liabilities or undisclosed financial maneuvers (“Directive 2011/61/EU”, 2011).

Financial Integrity and Stability

Articles 30(1) and 30(2) impose restrictions on AIFMs to protect the financial stability and integrity of the companies they control. These articles prohibit AIFMs from facilitating or supporting the distribution of capital, reduction of capital, redemption, or acquisition of shares by the company within a specified period. These restrictions prevent asset stripping and ensure that the company's valuation remains intact, reducing the likelihood of mispricing at the time of IPO ("Directive 2011/61/EU", 2011).

Enhanced Annual Reporting

Article 29 requires AIFMs to ensure that annual reports for non-listed companies they control provide a true and fair view of the company's financial situation. This includes disclosing significant events post-financial year, future developments, and information regarding the acquisition of own shares. Additionally, Article 30(3) mandates that information about the control and financial health of the company must be promptly communicated to employees or their representatives. These requirements ensure that all stakeholders are well-informed about the company's financial health, contributing to more accurate pricing ("Directive 2011/61/EU", 2011).

In summary, AIFMD's comprehensive regulatory framework, focusing on transparency, detailed reporting, financial stability, and stakeholder communication, is highly relevant to venture capital involvement in IPOs. Where the implementation of the AIFMD regulatory changes in 2011 helps create a more transparent and stable market environment than during the market crash in 2008, reducing the likelihood of mispricing. This makes the AIFMD an ideal regulatory shock to study using a differences-in-differences regression approach, as it provides a clear before-and-after scenario with a clear control group of non-VC backed-IPOs.

5 Descriptive Analysis

Data Analysis Overview

In this section, we will provide an in-depth overview of our merged and cleaned dataset, examining its key characteristics and distributions. This preliminary analysis serves as a foundation for further statistical tests. Our goal is to gain a clear understanding of the data to ensure its usability for the main analysis. We present descriptive analyses by summarizing key data attributes and highlighting preliminary findings through summaries and graphical representations.

Overview of Descriptive Statistics

From the refined dataset, we have a complex collection of 6404 initial public offerings, of which 210 are supported by venture capital, with data extending back to the year 2000. The United Kingdom (GB) emerges as the leading country for the highest number of IPOs, covering both venture capital-backed and non-venture capital-backed entities. For first-day returns, the technology sector displays the highest average performance. These findings are in line with those of Bessler and Seim (Bessler & Seim, 2012).

In Table 5.1, we present the annual averages of the first-day return, distinguishing between those with venture capital backing and those without. It is observable that during periods characterized by a high frequency of IPOs, the mispricing for VC backed firms tends to have a greater average than when low frequency.

Similar to other studies, we observe a significant difference in average first-day returns between venture capital backed and non-VC backed IPOs. Specifically, we find an average first-day return of 18.021% for VC-backed firms and 11.62% for non-VC backed firms.

Ritter’s findings from 2023 conducted an analysis of 9,089 firms from the period of 1980 to 2021 revealed an average return of 13.5% for non-VC backed IPOs, while the subset of 3,659 VC-backed firms demonstrated a significantly higher average first-day return of 27.0% (Ritter et al., 2023). Our findings for non-VC-backed IPOs in Europe align more closely with Ritter’s findings than those for VC-backed IPOs in Europe, which could indicate a greater degree of underpricing in the U.S. market for VC-backed IPOs. The visualization of the distribution of first-day returns over time, categorized by investment type, is presented in Figure 5.1.

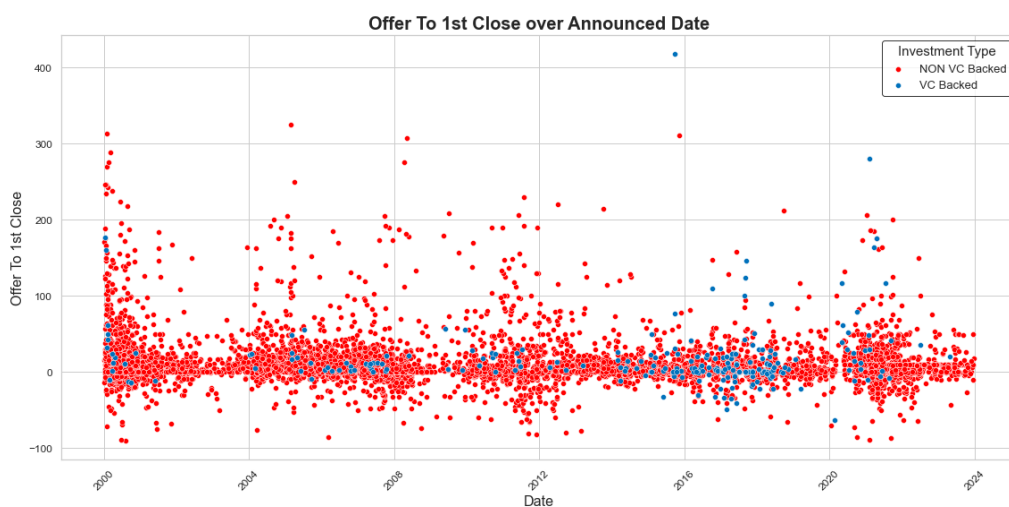


Figure 5.1: Distribution of the first day return

First day returns

Based on the initial offering price, the average first-day return for each year, categorized by investment type is presented in Table 5.1. It is evident from the table that the average first-day returns are highly volatile and exhibit significant fluctuations over the observed period for both groups.

It is clear that most companies filing for IPO experience an appreciation in their share price compared to the offer price of the IPO. This is in line with the phenomenon of underpricing, as discussed in the literature review. Some notable IPOs during the research period include Zalando, HelloFresh, Just Eat, and Blue Prism which all had a positive first day return.

Year	IPOs	FDMR Non-VC	FDMR VC	% FDMR Tech
2000	787	22.02%	38.41%	27.67%
2001	234	9.81%	-11.07%	4.35%
2002	133	8.08%	0.00%	8.21%
2003	103	12.25%	0.00%	5.85%
2004	318	15.77%	16.59%	10.03%
2005	456	15.58%	21.06%	12.89%
2006	536	9.44%	4.36%	9.49%
2007	482	10.00%	8.46%	7.69%
2008	205	9.28%	21.88%	-5.79%
2009	60	16.31%	29.34%	46.00%
2010	234	11.04%	17.57%	8.87%
2011	310	12.29%	15.96%	22.57%
2012	170	11.38%	7.01%	19.72%
2013	172	10.32%	8.85%	17.10%
2014	251	6.55%	6.84%	8.64%
2015	255	7.67%	30.34%	7.22%
2016	222	5.48%	8.24%	9.11%
2017	299	7.55%	10.13%	12.34%
2018	230	5.00%	7.48%	3.39%
2019	132	10.02%	-2.48%	17.64%
2020	179	12.32%	27.75%	14.00%
2021	572	10.00%	75.75%	8.12%
2022	178	5.72%	35.07%	-0.81%
2023	119	9.61%	20.53%	21.40%

Table 5.1: Average First-Day Returns of Venture Capital (VC) and Non-Venture Capital (Non-VC) IPOs, Including Technology Sector, from 2000 to 2023

Countries

To highlight country-specific IPO patterns, we used the bar charts below. For VC-Backed IPOs, Sweden tops the list with 63 IPOs, followed by the United Kingdom with 49 IPOs, and France with 34 IPOs. The remaining IPOs are distributed across other countries. This distribution is illustrated in Figure 5.2.

The average first-day returns was 16.59% for Sweden, 32.39% for the United Kingdom, and 6.64% for France, with the United Kingdom showing the highest first-day returns among countries with more than three IPOs.

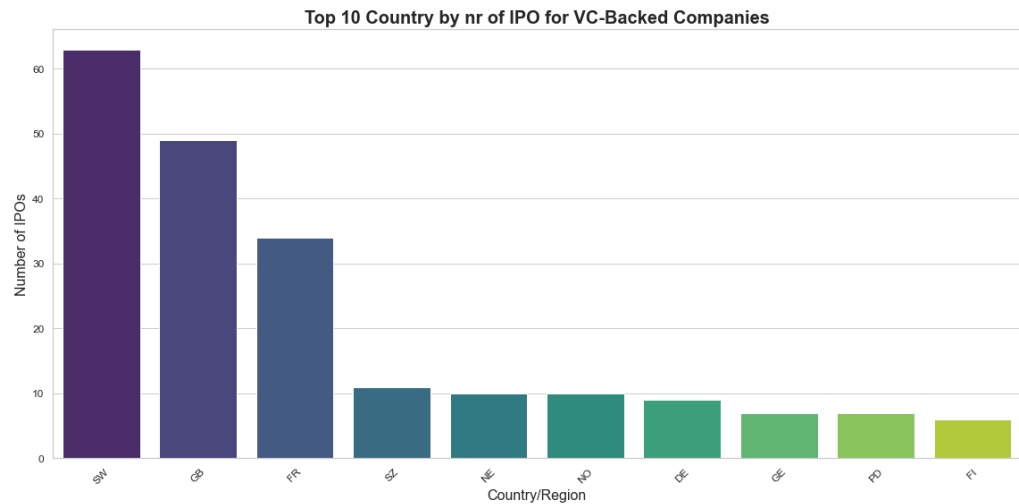


Figure 5.2: IPOs per Country for VC-Backed Companies

For non-VC-backed companies, the United Kingdom (GB) has the highest number of IPOs with 1,758, followed by Poland (PD) with 733, and France (FR) with 544. The remaining IPOs are distributed across various other countries, as shown in figure 5.3. Average first-day returns for these countries are 16.07% for the United Kingdom, 17.91% for Poland, and 5.29% for France, with Belgium exhibiting the highest first-day returns on average.

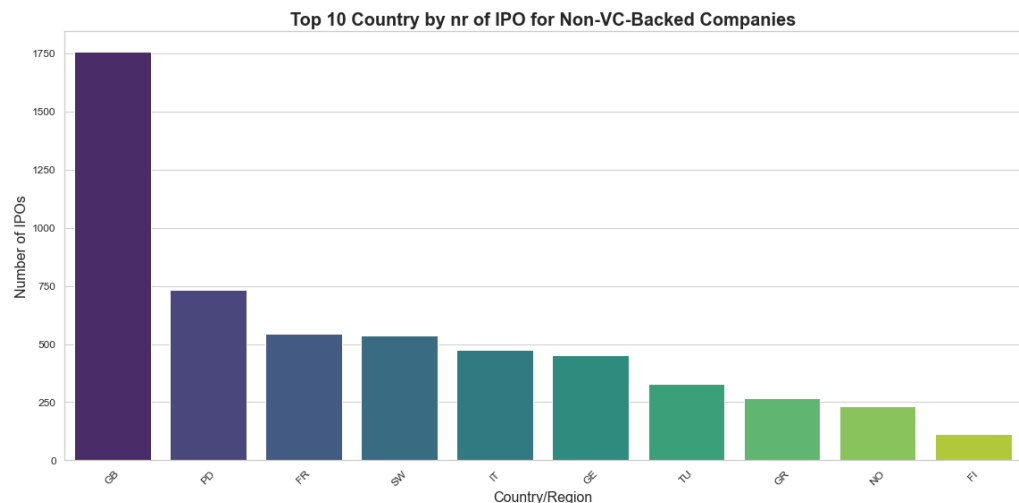


Figure 5.3: IPOs per Country for Non-VC-Backed Companies

Industrie Group

The Bloomberg Industry Classification System (BICS) provides a detailed framework for categorizing companies into specific sectors, assisting in the analysis of industry-specific trends (Bloomberg, 2023). For the Pichoobk data, we used the industry categorizations provided in their database. In our dataset of VC-backed companies, the top three sectors with the most IPOs are Biotech with 37 IPOs, Software with 30 IPOs, and Healthcare-Products with 18 IPOs. The remaining IPOs are distributed across other sectors.

For non-VC-backed companies, the top three sectors with the most IPOs are software with 494 IPOs, commercial services with 365 IPOs, and Financial services with 311 IPOs, leaving the remaining IPOs in other sectors. This distribution is visualized in figure 5.3. The tech sector displays the highest average first-day returns for both VC and Non-VC-backed companies, which is consistent with the findings of Bessler and Seim (Bessler & Seim, 2012).

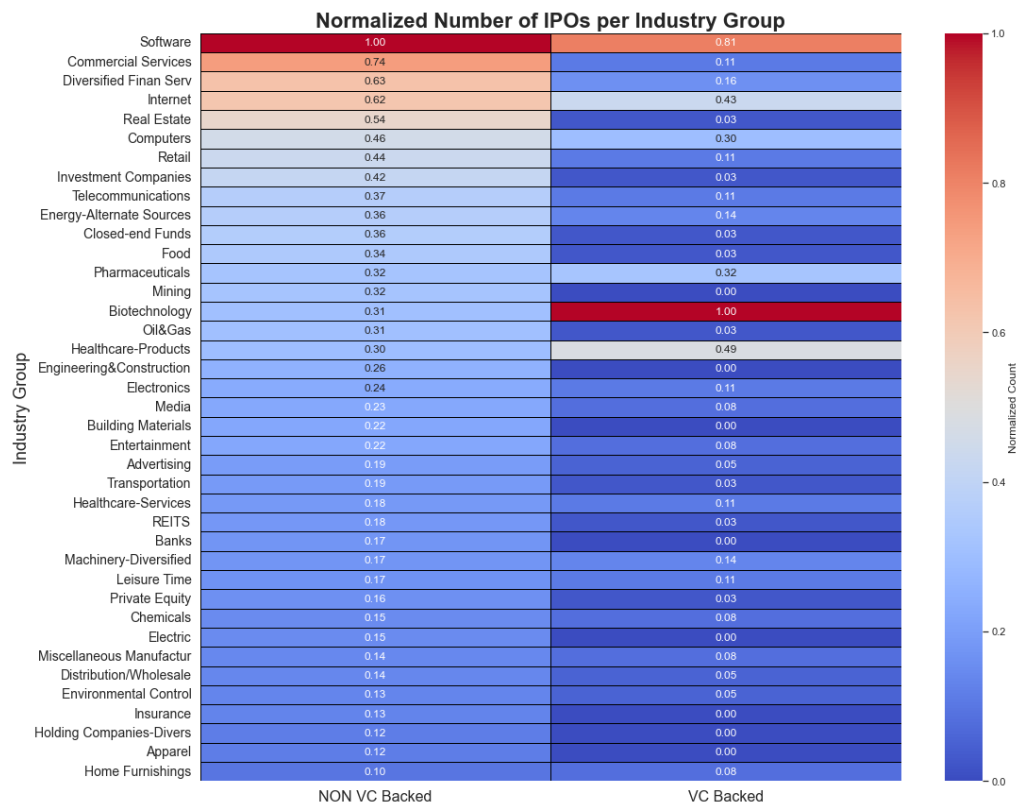


Figure 5.4: Heatmap distribution VC and Non-VC backed IPOs in BICS Industry Group

Offer Size

The offer size of an IPO is determined by multiplying the number of shares offered to the market by the offer price per share. Figure 5.4 displays the distribution of offer sizes in our dataset. The non-VC-Backed chart shows that the majority have IPOs with an offer size of less than \$50 million, followed by the \$50-150 million range. Notably, there is a significant number of observations for offer sizes above \$750 million.

The VC-backed chart indicates that these companies primarily raise \$50 million or less, with a significant drop in frequency as the offer size increases. However, there is a notable spike for offer sizes greater than \$750 million. This distribution reveals that both non-VC-backed and VC-backed companies tend to have smaller offer sizes, with a few instances of very high offer sizes.

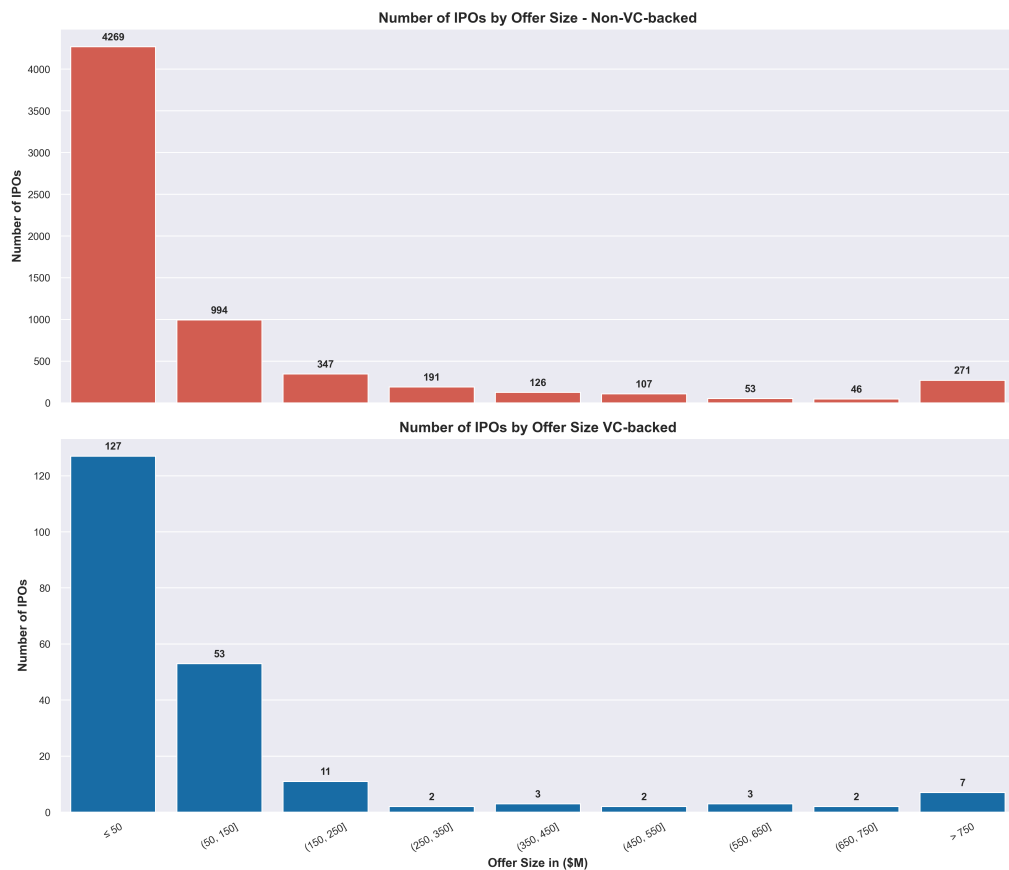


Figure 5.5: Number of IPOs by offer size in million USD: Non-VC backed (Red) vs. VC backed (Blue)

6 Results and analysis

In this section, we will conduct a detailed analysis to address our hypotheses and display the corresponding findings. We will use both non-statistical and statistical methods. Non-statistical analysis will help to identify obvious patterns in the data, while statistical analysis will determine if these patterns are significant.

6.1 VC-Backed IPOs Show Positive FDR

For the first hypothesis, we aim to test whether the mean first-day return for VC-Backed IPOs is significantly greater than zero. To achieve this, we will use both non-parametric and parametric statistical tests. To assess the normality of the data, we conducted a Shapiro-Wilk test. The test resulted in a very low p-value, indicating that the data does not follow a normal distribution.

Since the data does not follow a normal distribution, we used the Wilcoxon signed-rank test to assess the significance of the first-day return against zero. The p-value of 1.02×10^{-15} strongly supports rejecting the null hypothesis that the return is zero or negative. Consequently, this result confirms that the median first-day return for VC-backed IPOs, measured at 6.59%, is significantly greater than zero with a high level of statistical confidence.

We conducted a one-sample t-test to compare the sample mean to a hypothesized mean of zero. With a sufficiently large sample size, the distribution of the sample mean approximates normality, regardless of the population's distribution (Kwak & Kim, 2017). The t-test yielded a one-tailed p-value of 4.2×10^{-9} . Despite previous signs of non-normality, this extremely low p-value allows us to reject the null hypothesis that the mean return is zero or negative. This suggests that the first-day return is significantly greater than zero with 95% confidence.

Both parametric and non-parametric tests consistently support the hypothesis that the first-day returns of VC-backed IPOs are significantly positive. Despite the initial non-normality indicated by the Shapiro-Wilk test, the application of the Central Limit Theorem (CLT) and a large sample size justified the use of the t-test. The results from the t-test confirmed the findings of the non-parametric Wilcoxon signed-rank test. This analysis provides evidence that VC-backed IPOs tends to have positive returns on their first trading day.

6.2 European VC-Backed IPOs Steal the Show

In the second hypothesis, we investigate whether European VC-backed IPOs exhibit higher first-day returns compared to non-VC-backed IPOs. We initiated our analysis with a Levene's test, yielding a highly significant p-value of 0.0015. This result underscores that the variances in first-day returns between VC-backed and non-VC-backed IPOs are significantly different.

Given this significant difference in variances, we used Welch's t-test to compare the mean returns. Welch's t-test is appropriate in this context as it does not assume equal variances and is suitable under conditions of heteroscedasticity.

The Welch's t-test produced a one-tailed p-value of 0.0014. Since this p-value is below the threshold for statistical significance at the 95% confidence level ($p < 0.05$), it indicates a statistically significant difference in mean first-day returns. This finding suggests that VC-backed IPOs outperformed non-VC-backed IPOs in Europe from 2000 to 2023. Leading us reject our H_0 , thereby validating:

$$H_1 : \bar{R}_{VC} > \bar{R}_{NVC}$$

To further validate our findings from the t-test, we conducted a non-parametric analysis using the Mann-Whitney U test. Unlike the Welch's t-test, this test does not assume normality and it's is used to determine whether one of two independent samples tends to have larger values than the other. The Mann-Whitney U test returned a p-value of 0.0062, indicating

that the median first-day return for VC-backed IPOs is significantly greater than that for non-VC-backed IPOs. This result aligns with our t-test findings, providing robust support for our hypothesis.

Findings

Our analysis robustly supports the beneficial impact of venture capital backing on IPO performance in Europe. For the first hypothesis, both the Wilcoxon signed-rank test and the one-sample t-test confirmed that VC-backed IPOs have significantly positive first-day returns. These results are consistent with previous findings, such as those reported by Ritter, Bessler and Seim (Ritter et al., 2023) (Bessler & Seim, 2012).

For the second hypothesis, Welch's t-test and the Mann-Whitney U test both demonstrated that VC-backed IPOs have significantly higher first-day returns compared to non-VC-backed IPOs. Levene's test justified the use of Welch's t-test due to significant differences in variances between the groups.

Overall, the consistent results from both parametric and non-parametric tests validate that European VC-backed IPOs outperform non-VC-backed IPOs in terms of first-day returns. These results supports the idea that venture capital provides a quality signal to the market, aligning with theory, which suggests that investor's view VC-backed firms as a premium. these aligns with Barry's findings from 1990 and Lee's from 2004 (Barry et al., 1990) (Lee & Wahal, 2004). Both studies observed higher returns for VC-backed firms, reinforcing the positive impact of VC involvement on IPO performance.

On the other hand, it could be argued that the higher first-day returns of VC-backed IPOs might be influenced by other factors such as the broader economic environment or industry-specific trends. The high first-day returns may be related to market hype, overoptimism, and temporary underpricing, or "fads," as defined by (Ritter, 1991). The market timing also plays a role, as companies may choose to go public during periods of high investor sentiment and enthusiasm. A longer time frame could potentially reveal if VC-backed companies add additional value over time.

6.3 Hypothesis 3

After establishing the regulatory relevance of AIFMD in the methodology section, we demonstrated that the 2011 regulatory change impacts only the observation group, leaving the control group unaffected. This sets the stage for testing Hypothesis 3. This hypothesis serves as a validation measure. If the data does not show statistical significance, it indicates that the regulation does not affect the treatment group. Additionally, we used the Mann-Whitney U test and a t-test to further investigate any potential statistical significance in the data ⁶.

6.3.1 Statistical Analysis and Key Findings

The t-test and the non-parametric Mann-Whitney U test revealed a statistically significant difference at the 95% confidence level between the pre-law and post-law periods for VC-backed companies, but not for non-VC-backed companies. This could indicate that the regulatory change significantly impacted VC-backed companies, while non-VC-backed companies remained unaffected, aligning with our hypotheses. (Appendix table A.3)

Since Levene's test indicates that the variances are significantly different, suggesting the presence of heteroscedasticity in the data, we used robust standard errors to account for this.

The DID model provides insights into the impact of the regulatory change on the first-day return for VC-backed companies. The model's R-squared value of 0.002 indicates that it explains a very small portion of the variance in first-day returns. While this may suggest that other factors not included in the model may be influencing the dependent variable, the individual findings still offer valuable insights.

The F-statistic of 2.794 and a corresponding p-value of 0.0392 suggest that the

⁶All IPOs for both datasets are included up to June 8, 2011, when the law was confirmed. No significant outliers in terms of underpricing or the number of IPOs were observed immediately before or after the law's confirmation.

overall model is statistically significant at the 5% level, meaning it has some explanatory power. However, the coefficients of the explanatory variables provide nuanced information. The constant term is 12.7956, indicating that on average, the first-day return was approximately 12.8% for non-VC-backed firms.

The coefficient for the PostLaw variable is -2.7499, but it is not statistically significant ($p = 0.198$), suggesting that the regulatory change did not significantly affect the first-day returns for non-VC-backed firms. The VC-Backed variable has a coefficient of 7.3244, indicating an average first-day return of 20.12% for VC-backed firms. This result is statistically significant at the 90% confidence level, though it does not meet the conventional 95% confidence threshold, suggesting that the VC effect during the research period was somewhat smaller than typically observed.

6.3.2 Impact on VC-Backed Companies

The interaction term (PostLaw*VCBacked) has a coefficient of -10.2003 and is borderline significant ($p = 0.060$). This suggests that the regulatory change may have reduced the first-day return for VC-backed firms by approximately 10.2 percentage points compared to non-VC-backed firms. This effect is noteworthy and warrants further investigation despite the p-value being slightly above the conventional 0.05 threshold (Appendix table A.4).

6.3.3 Interpretation and Implications

The findings from our Difference-in-Differences model provide important insights into the impact of the 2011 regulatory change on IPO outcomes, particularly for VC-backed companies. By employing a robust statistical framework that includes heteroscedasticity-robust standard errors, we ensured the reliability of our results, given the heteroscedasticity indicated by Levene's test.

Our analysis indicates that the regulatory change, PostLaw, is not statistically significant, suggesting that the law did not have a significant

direct impact on non-VC-backed firms, which aligns with our hypothesis. The mean return during this period also aligns closely with the findings from 2023 by Ritter, showing a 0.7% difference in mean return for non-VC-backed IPOs (Ritter et al., 2023). The results also demonstrate underpricing beforehand, with VC-backed firms experiencing higher first-day returns compared to non-VC-backed firms, although this result is significant only at the 90% confidence level.

The interaction term (PostLaw*VCBacked) is of particular interest. With a coefficient of -10.2003 and a borderline significance ($p = 0.060$), it could suggest that the regulatory change may have reduced the first-day return for VC-backed firms by approximately 10.2 percentage points relative to non-VC-backed firms. This finding implies that the regulatory change specifically affected VC-backed firms, potentially by imposing more strict disclosure and transparency requirements, thereby reducing the informational asymmetry and speculative behavior often associated with these firms. As visually shown in Figure 6.2, there is a clear reduction in underpricing after 2011 compared to non-VC-backed IPOs.

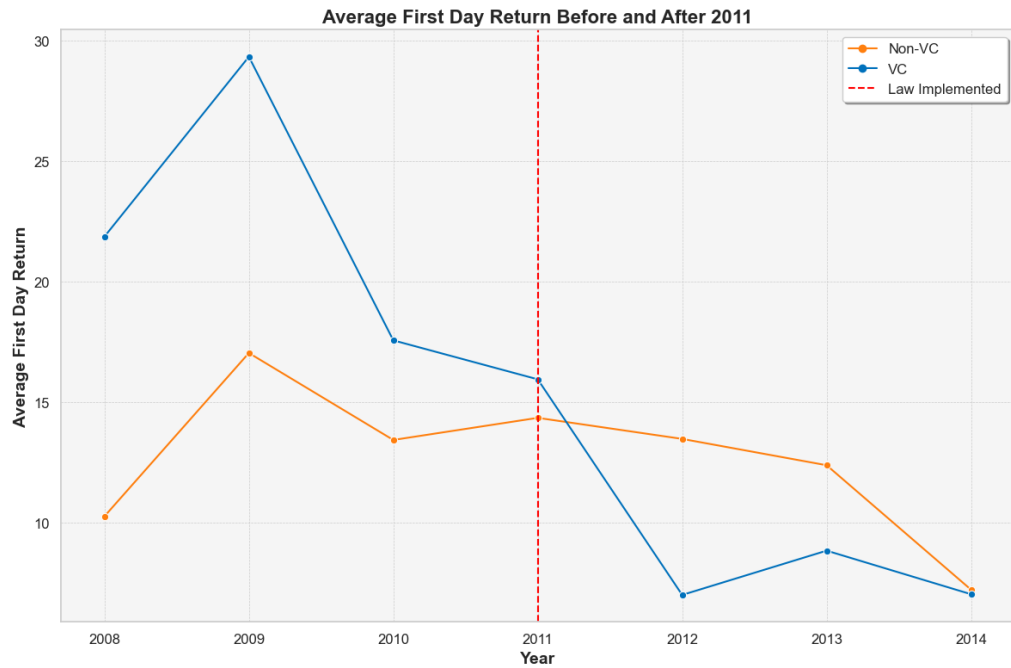


Figure 6.1: DID Model on average returns, three years pre and post law.

The decrease in first-day returns for VC-backed firms post-regulation could indicate that such measures can mitigate some asymmetric information in the financial market. These findings contribute to the broader understanding of how regulatory changes have led to greater market efficiency, compelling VCs to disclose information that was not previously common knowledge. Despite this, a visible difference in first-day returns between VC-backed and non-VC-backed firms persists even after the IPO. Additionally, both the t-test and the Mann-Whitney U test validated our findings, showing a statistically significant difference between the pre-law and post-law periods for VC-backed companies.

The results could indicate important implications for policymakers, investors, and market participants. For policymakers, the findings could suggest that regulatory frameworks such as represented by AIFMD might enhance market efficiency and transparency, particularly for VC-backed firms. The significant reduction in first-day returns for these firms post-regulation could imply that targeted measures may address market inefficiencies.

For investors, particularly those involved in VC-backed IPOs, the results indicate a reduction in first-day returns post-regulation, suggesting less overpricing and speculative trading. However, the low explanatory power of the model means investors should also consider other market factors in their decisions. For market participants, understanding the specific impact of regulations on different types of firms can improve strategic planning and decision-making. The observed reduction in first-day returns for VC-backed firms underscores the importance of considering a range of market indicators when assessing IPO investments.

In conclusion, while the 2011 regulatory change appears to have reduced first-day returns for VC-backed firms, the low explanatory power of the model indicates that other factors also influence IPO outcomes.

6.4 Hypothesis 4

Impact of Legal Origins on Market Efficiency

Building on La Porta's study on investor protection, this hypothesis examines the impact of the AIFMD regulation on venture-backed IPOs across various European countries, classified by their legal origins. By leveraging differences in investor protection and enforcement quality, we aim to assess the AIFMD's regulatory effect on market efficiency and IPO underpricing across different legal systems. We expect regions with higher market efficiency to experience significantly lower levels of underpricing compared to regions with lower market efficiency.

We classified 17 European countries into four legal origins: Common Law, French Civil Law, German Civil Law, and Scandinavian Law. By using this classification, we applied the same criteria as in Hypothesis 3, comparing IPO underpricing three years before and after 06.08.2011. This approach allows us to understand how the effectiveness of the AIFMD regulation varies across different legal systems. In the next section, we will present the findings for the four different legal origins.

6.4.1 French Civil Law Countries

The analysis of French Civil Law countries, which Porta suggests have the weakest investor protections, is expected to reveal minimal differences in underpricing following the implementation of the law ("Directive 2011/61/EU", 2011). As illustrated in Figure 6.2, the results indicate an small increase in venture capital-backed underpricing after June 8, 2011, whereas the broader market experienced a decline.

From the analysis, we observe that the constant term is 5.4702, which is significant at the 1% level. This indicates a baseline level of underpricing for non-VC-backed IPOs. The coefficient for the post-law period is -2.27, which is not statistically significant ($p = 0.159$), indicating no significant change in the underpricing for non-VC backed IPOs after law.

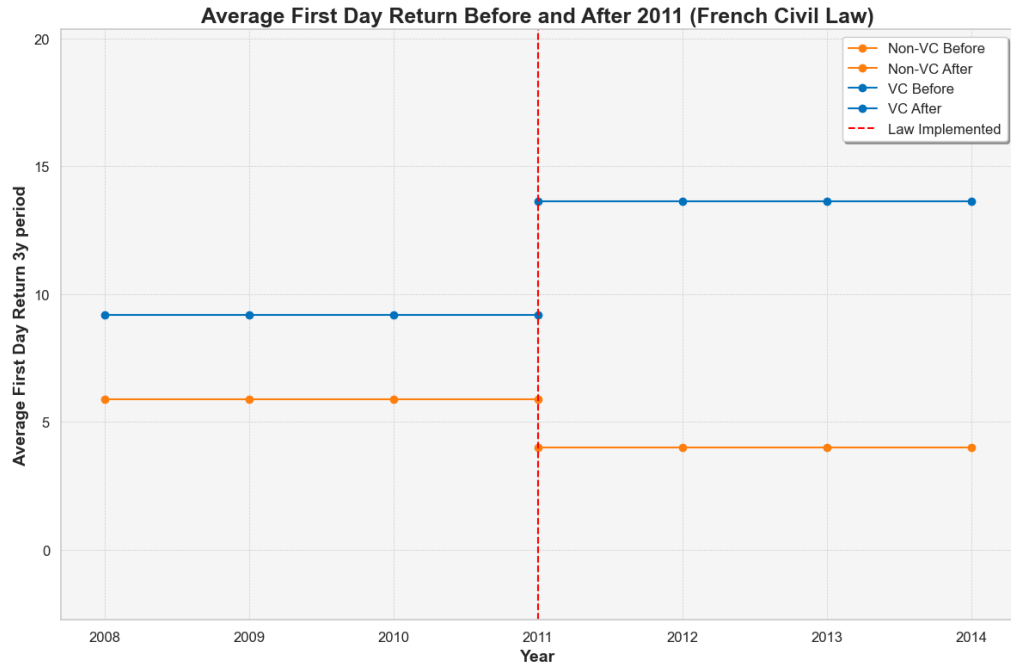


Figure 6.2: AFDR, 3y average span, French Civil Law

The coefficient for VC-backed companies (VCBacked) is 6.7176, which is close to significance at the 10% level ($p = 0.124$). This indicates that VC-backed IPOs did not experience significantly underpricing compared to non-VC-backed IPOs before the law change, even though the result is close to the threshold for significance. The interaction term (PostLaw*VCBacked) has a coefficient of 0.6474, not statistically significant ($p = 0.950$), indicating the law did not significantly impact the first-day returns for VC-backed IPOs.

The model's R-squared value of 0.016 indicates that it explains only a very small portion of the variance in first-day returns. The F-statistic, with a p-value of 0.0913, suggests that the overall model is not statistically significant at the 5% level but is at the 10% level. These findings indicate that the implementation by AIFMD did not lead to a significant reduction in the first-day returns, particularly for VC-backed companies. This result aligns with the expectation that French Civil Law countries, which generally provide weaker investor protections, would exhibit less responsiveness to the regulatory changes introduced by AIFMD (Appendix table A.5).

The empirical evidence does not provide strong support for Hypothesis 4, as the model is not statistically significant. As expected, there was minimal difference in underpricing for VC-backed IPOs following the AIFMD regulatory implementation. However, since the model shows little explanatory power, indicating that other factors may be influencing the underpricing beyond the variables included in this analysis, we cannot definitively say that this part of hypothesis 4 is supported.

When the explanatory variables in the regression model are not statistically significant but show substantial differences in their coefficients, it could indicate that while there might be a meaningful size effect, there is also a high degree of uncertainty or variability associated with the estimate. This can result from factors such as a small sample size or low statistical power, leading to less precise estimates in the Difference-in-Differences analysis. In our results, the non-significance of the coefficients for PostLaw, VCBacked, and PostLaw*VCBacked suggests that there is high variability in our estimates. This indicates that while there may be an effect, our study lacks the precision to confidently identify it, likely due to these factors.

6.4.2 German Civil Law Countries

Continuing with the impact from the regulatory changes presented by AIFMD on IPO underpricing in German Civil Law countries, these countries are expected to provide intermediate levels of investor protection and market efficiency. As shown in Figure 6.3, both the observation group and the treatment group experienced a decrease in first-day returns. However, the treatment group saw a greater reduction, leading to less underpricing after the implementation of the law.

The results for German civil law countries shows that the constant term is 19.4820, significant at the 1% level, indicating underpricing before the law. The coefficient for the post-law period (PostLaw) is -0.6578, suggesting no significant change in underpricing for non-VC-backed IPOs following the legal implementation.

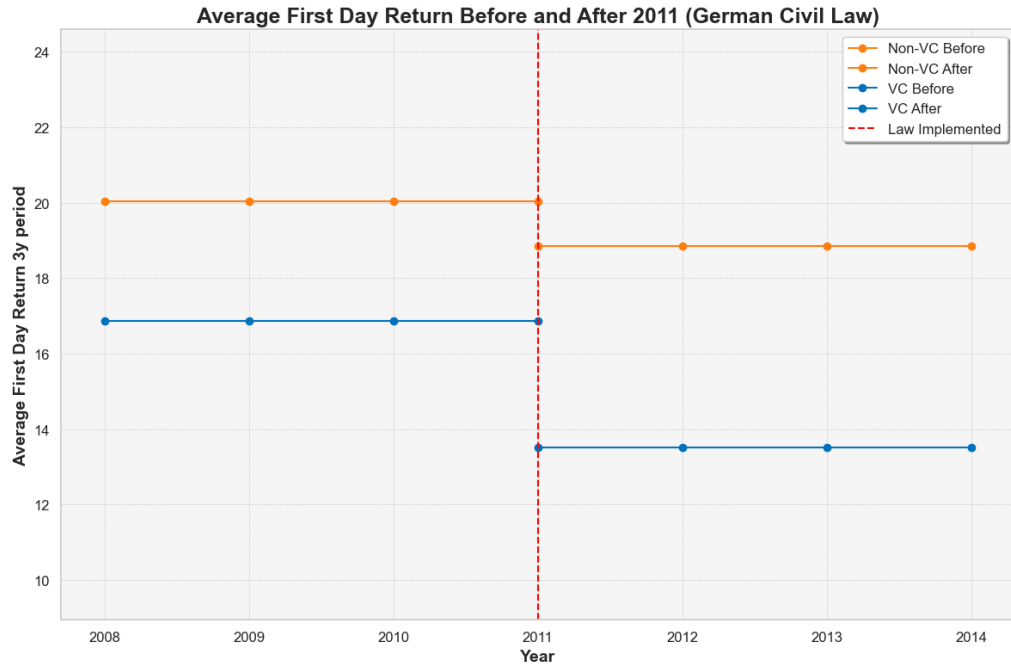


Figure 6.3: AFDR, 3y average span, German Civil Law

For VC-backed companies (VCBacked), the coefficient is -1.6698, indicating no significant difference in underpricing for VC-backed IPOs compared to non-VC-backed IPOs before the law. The interaction term (PostLaw*VCBacked) has a coefficient of -14.1294 ($p = 0.277$), suggesting no significant reduction in underpricing for VC-backed IPOs post-AIFMD. Despite the large reduction in underpricing indicated by the coefficient, the lack of statistical significance means this result should be interpreted with caution.

The F-statistic of 0.8223 ($p = 0.482$) suggests that the overall model is not statistically significant at the 5% level. Additionally, the R-squared value is very low, indicating that the model explains nearly none of the variance in the dependent variable. These findings indicate that the regulatory implementation by AIFMD did not significantly impact the underpricing in German Civil Law countries, particularly for VC-backed IPOs (Appendix table A.5).

6.4.3 Common Law Countries

Common law countries typically provide the strongest investor protections. As illustrated in figure 6.4, VC-backed companies exhibited higher levels of underpricing compared to the control group. However, this underpricing is dramatically reduced after the law in 2011, while the control group remained relatively unchanged, confirming our expectations on the visual data.

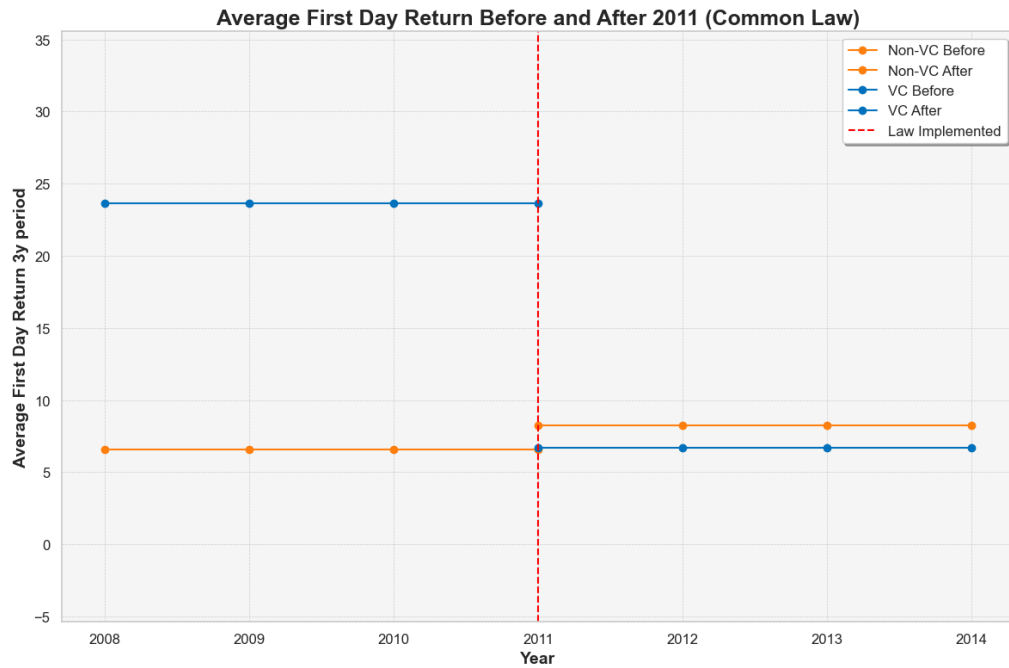


Figure 6.4: AFDR, 3y average span, Common Law

For non-VC-backed IPOs, the average underpricing was 5.9923, statistically significant at the 1% level ($p < 0.001$). After the law, the underpricing increased by 2.9838 percentage points, although this was only significant at the 10% level ($p = 0.100$). For VC-backed IPOs, the average underpricing was 17.6231 ($p = 0.006$), indicating significantly higher underpricing compared to non-VC-backed IPOs before the law, significant at the 1% level. VC-backed IPOs experienced a substantial reduction in underpricing by -19.8749 ($p = 0.004$), also significant at the 1% level.

This significant reduction, observed only in VC-backed IPOs, suggests that the regulation had a more pronounced impact on these companies, highlighting the potential causal effect of venture capital backing on the

response to regulatory changes. However, the model's R-squared value of 0.022 indicates that it only explains a small portion of the variance in the dependent variable. The F-statistic of 3.067 ($p = 0.0285$) shows that the overall model is statistically significant (Appendix table A.5).

Empirical evidence from common law countries provides partial support for Hypothesis 4. While the law did significantly reduce underpricing for VC-backed IPOs, indicating some regulatory impact, the overall explanatory power of the model is weak. This suggests that other factors may also be at play and that the regulatory framework, although important, might not be the sole or dominant factor influencing the reduction in underpricing. The statistical significance of these findings highlights the necessity for further research to identify other contributing factors.

6.4.4 Scandinavian Civil Law Countries

Our analysis concludes with the Scandinavian civil law countries, known for providing strong protections and high enforcement quality. Similar to figure 6.4 above, figure 6.5 display that VC-backed underpricing is dramatically reduced after the law in 2011, while the control group remained relatively unchanged, again confirming our expectations on the visual data, while the statistical interpretation the results looks a bit different.

The results suggest that AIFMD's regulatory implementation did not significantly impact underpricing for VC-backed IPOs. The significant coefficient for VCBacked indicates that VC-backed IPOs generally had larger underpricing before the law at 31.13%. However, the interaction term (PostLaw*VCBacked) and the overall model are not statistically significant, suggesting that the regulatory changes did not have the intended effect. The very low R-squared value indicates that other factors not included in the model influencing the underpricing.

These findings highlight the complexity of the market and suggest the need for further research to identify additional factors. Although the data show a -37.3065 percentage point decrease in underpricing after the

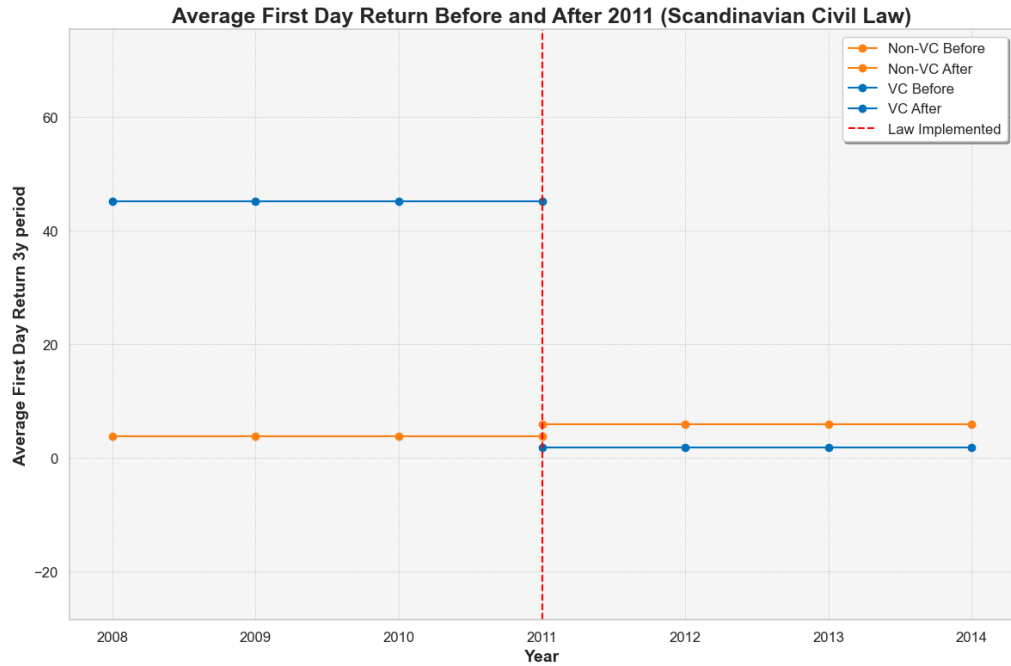


Figure 6.5: AFDR, 3y average span, Scandinavian Civil Law

legal implementation for VC-backed IPOs, this decrease is not statistically significant. As a result, we do not have enough evidence to conclusively support our hypothesis (Appendix table A.5).

Conclusion hypothesis 4

The findings reveal interesting trends that align with our expectations. They suggest that higher-governance legal regions would experience significantly lower levels of underpricing after the regulatory implementation by AIFMD. However, when examining the empirical evidence, it becomes clear that there is insufficient evidence to conclude a significant difference between legal origins due to the lack of explanatory power in the models across all regions.

For all regions except German Civil Law, there was a significant difference in the average first-day return between non-VC-backed and VC-backed IPOs during the research period, which is consistent with our findings in Hypothesis 2. While we anticipated a substantial reduction in underpricing for Common Law and Scandinavian Civil Law regions, the observed decreases were not statistically significant. Common Law was the only legal region

where the post-law change exhibited a statistically significant difference. This aligns with La Porta's findings that countries with stronger legal protections for investors typically have more advanced capital markets, positively impacting the performance of venture-backed IPOs (Porta et al., 1998) (Bessler & Seim, 2012).

In conclusion, our findings for hypothesis 4 did not provide sufficient evidence to support the hypothesis, as the statistical analysis did not reveal a significant impact. This suggests that the AIFMD's regulatory implementation did not lead to a significant change in underpricing across different legal origins. There are, however, small indications that the regulatory implementation could have had a small impact on reducing IPO underpricing in higher-governance regions as seen in the common law region. Despite these indications, the lack of statistical significance prevents us from conclusively supporting hypothesis 4. These findings highlight the complexity of market dynamics and suggest the need for further research to identify additional factors that influence IPO performance.

7 Conclusion

Our study aimed to investigate the potential relationship between venture capital backing and IPO underpricing in European markets from 2000 to 2023. We examined this relationship by investigating: (1) whether VC-backed IPOs yield positive first-day returns, (2) whether venture capital backed IPOs systematically outperformed non-VC-backed IPOs, (3) whether IPOs with VC backing showed less underpricing post-law compared to the control group during the -3 to +3 year timeframe, and (4) how these effects differ across different legal regions. We tested these hypotheses using multiple Difference-in-Differences analyses, Mann-Whitney U tests, and other parametric t-tests.

Our findings confirmed that VC-backed IPOs exhibit positive first-day returns, aligning with our initial hypothesis. This result was statistically significant, indicating that VC backing plays a crucial role in enhancing IPO performance on the first day of trading. Furthermore, we observed a significant difference in underpricing between VC-backed and non-VC-backed IPOs, reinforcing the notion that venture capital positively influences IPO outcomes and confirming our second hypothesis.

Our study highlights the impact of the implementation of the Alternative Investment Fund Managers Directive (AIFMD) in 2011, which acted as a market shock, enabling us to compare the European market before and after this event. This analysis indicated that VC-backed IPOs experienced reduced underpricing post-law implementation. However, the low explanatory power of our Difference-in-Differences model suggests that, while there is a correlation, causality cannot be firmly established as other factors likely influenced the observed outcomes.

Building on this, we utilized the categorization of countries based on legal origin by Rafael La Porta. This allowed us to examine country-specific effects and determine if higher governance regions experienced significantly lower levels of underpricing compared to lower governance regions for VC-

backed companies. The model indicated slight evidence that new legal regulation may have reduced IPO underpricing, but the lack of statistical significance and low explanatory power prevented us from fully supporting this hypothesis.

Thus, our hypotheses are partially confirmed. The findings are relevant for academics and researchers, contributing to the literature on IPO underpricing and venture capital. The study offers insights into how regulatory changes could affect market dynamics, making it particularly relevant for researchers interested in law, finance, and market behavior.

For investors, the results emphasize the importance of VC backing in achieving higher first-day returns, suggesting that the presence of venture capital can be a positive signal when evaluating IPOs. For policymakers, the study highlights the impact of regulation in shaping market outcomes. The regulation introduction by AIFMD appears to have moderated extreme IPO underpricing to some extent, particularly in high-governance regions. Policymakers could use this information to refine regulations further, enhancing market stability and transparency.

Our findings align with existing literature, such as Bessler and Seim, which emphasizes the positive impact of venture capital on IPO performance. However, the regulatory introduction by AIFMD and its specific effects on IPO underpricing in different governance contexts adds a new dimension to the discussion.

While our study provides valuable insights, it has limitations. The low explanatory power of the DID model suggests that additional factors influencing IPO underpricing were not accounted for. Future research could incorporate more variables or explore different methodologies to enhance explanatory power. Additionally, extending the comprehensiveness of the data by including IPOs from other databases could improve the dataset, as there may be more VC-backed IPOs in Europe that were not accounted for.

Given our findings and the persistence of these trends throughout our research, one could expect future IPOs to achieve higher first-day returns if they are venture capital-backed. This underscores that there is a systematic effect of venture capital on company valuation prior to going public and indicates a correlation between VC backing and higher first-day returns compared to non-VC-backed companies.

Suggestions for Further Research

Based on our study, future research should considering distinguishing between established and non-established venture capital firms. To bring deeper insights into Lee and Wahal's grandstanding theory in Europe (Lee & Wahal, 2004). Future research could investigate how the experience, reputation, and resources of established VC firms compare to those of newer VCs, and how these factors affect IPO outcomes. This understanding could help identify best practices and strategies for successful IPOs.

Moreover, our study did not find any country that had previously implemented advanced laws similar to 32011L0061 by AIFMD, affecting the control group comparably. This lack of prior research underscores the significance of our findings and highlights a gap in the literature. Therefore, future research could benefit from exploring other regions or countries that may adopt similar regulations, enabling a comparative analysis to validate and extend our conclusions.

Integrating more variables into the models could enhance their explanatory power and increase R-squared values. Future research should consider including macroeconomic indicators, such as interest rates, as well as factors related to the venture capital firm. Additionally, using an extensive dataset of VC-backed IPOs, particularly from the 2011 timeframe, and increasing the sample size with other databases would enhance the statistical power and reliability of the findings, leading to more definitive conclusions.

In summary, expanding the scope of research to include a broader dataset and additional explanatory variables will likely yield a more comprehensive understanding of the factors influencing IPO performance in Europe, particularly in the context of regulatory changes with similar characteristics to the law presented by AIFMD in 2011. By addressing these areas, future research can build on our findings and contribute to a deeper understanding of the role of venture capital in the IPO process.

References

- ABOS. (2023, February). Correlation and causation | Australian Bureau of Statistics. Retrieved June 2, 2024, from <https://www.abs.gov.au/statistics/understanding-statistics/statistical-terms-and-concepts/correlation-and-causation>
- Alex Warfel & Caroline Suttie. (2023). National Venture Capital Association (NVCA). https://nvca.org/wp-content/uploads/2023/03/NVCA-2023-Yearbook_FINALFINAL.pdf
- Banton, C. (2023, May). Underwriter in Finance: What Do They Do, What Are Different Types? Retrieved June 2, 2024, from <https://www.investopedia.com/terms/u/underwriter.asp>
- Barry, C. B., Muscarella, C. J., Peavy, J. W., & Vetsuypens, M. R. (1990). The role of venture capital in the creation of public companies: Evidence from the going-public process [Place: Amsterdam Publisher: Elsevier B.V]. *Journal of financial economics*, 27(2), 447–471. [https://doi.org/10.1016/0304-405X\(90\)90064-7](https://doi.org/10.1016/0304-405X(90)90064-7)
- Bartolucci, F., Belotti, F., & Peracchi, F. (2015). Testing for time-invariant unobserved heterogeneity in generalized linear models for panel data. *Journal of Econometrics*, 184(1), 111–123. <https://doi.org/10.1016/j.jeconom.2014.09.002>
- Baum, J. A. C., & Silverman, B. S. (2004). Picking winners or building them? Alliance, intellectual, and human capital as selection criteria in venture financing and performance of biotechnology startups. *Journal of Business Venturing*, 19(3), 411–436. [https://doi.org/10.1016/S0883-9026\(03\)00038-7](https://doi.org/10.1016/S0883-9026(03)00038-7)
- Beatty, R. P., & Ritter, J. R. (1986). Investment banking, reputation, and the underpricing of initial public offerings [Place: Amsterdam Publisher: Elsevier B.V]. *Journal of financial economics*, 15(1), 213–232. [https://doi.org/10.1016/0304-405X\(86\)90055-3](https://doi.org/10.1016/0304-405X(86)90055-3)
- Berk, J., & DeMarzo, P. M. (2017). *Corporate finance* (Fourth edition.; Global edition.). Pearson.

- Bessler, W., & Seim, M. (2012). The performance of venture-backed IPOs in Europe [Place: London Publisher: Routledge]. *Venture capital (London)*, 14(4), 215–239. <https://doi.org/10.1080/13691066.2012.702447>
- Bloomberg, L. (2023). Classification data More precise and accurate industry classification data bloomberg. Retrieved June 12, 2024, from https://data.bloomberglp.com/professional/sites/10/131915_CDS_REF_Classification_SFCT_180315_DIG.pdf
- Bradley, Jordan, Ritter, J., & Wolf. (2004). The IPO quiet period revisited. *Journal of Investment Management*, 3–13. <https://joim.com/wp-content/uploads/emember/downloads/a0041.pdf>
- Brown, S. J., & Warner, J. B. (1985). Using daily stock returns: The case of event studies. *Journal of Financial Economics*, 14(1), 3–31. [https://doi.org/10.1016/0304-405X\(85\)90042-X](https://doi.org/10.1016/0304-405X(85)90042-X)
- Chambers, D., & Dimsion, E. (2009). IPO Underpricing over the Very Long Run [Place: Malden, USA Publisher: Blackwell Publishing Inc]. *The Journal of finance (New York)*, 64(3), 1407–1443. <https://doi.org/10.1111/j.1540-6261.2009.01468.x>
- Directive 2011/61/EU. (2011, June). Retrieved June 12, 2024, from <http://data.europa.eu/eli/dir/2011/61/oj/eng>
- Ewens, M., & Malenko, N. (2020, September). Board Dynamics over the Startup Life Cycle. <https://doi.org/10.3386/w27769>
- Fama, E. F. (1970). EFFICIENT CAPITAL MARKETS: A REVIEW OF THEORY AND EMPIRICAL WORK [Place: Oxford, UK Publisher: Blackwell Publishing Ltd]. *The Journal of finance (New York)*, 25(2), 383–417. <https://doi.org/10.1111/j.1540-6261.1970.tb00518.x>
- Fishe, R. P. H., Haynes, R., & Onur, E. (2019). Anticipatory Traders and Trading Speed [Place: Seattle Publisher: Cambridge University Press]. *Journal of financial and quantitative analysis*, 54(2), 729–758. <https://doi.org/10.1017/S0022109018000832>
- Gompers, P. A. (1996). Grandstanding in the venture capital industry [Place: Amsterdam Publisher: Elsevier B.V]. *Journal of financial economics*, 42(1), 133–156. [https://doi.org/10.1016/0304-405X\(96\)00874-4](https://doi.org/10.1016/0304-405X(96)00874-4)

- Gompers, P. A., & Lerner, J. (2003). The Really Long-Run Performance of Initial Public Offerings: The Pre-Nasdaq Evidence [Publisher: [American Finance Association, Wiley]]. *The Journal of Finance*, 58(4), 1355–1392. Retrieved June 2, 2024, from <https://www.jstor.org/stable/3648214>
- Johannessen, A., Christoffersen, L., & Tufte, P. A. (2021). *Introduksjon til samfunnsvitenskapelig metode* (6. utgave Asbjørn Johannessen, Per Arne Tufte og Line Christoffersen.). Abstrakt forlag. Retrieved January 14, 2024, from https://urn.nb.no/URN:NBN:no-nb_pliktpmonografi_000016149
- Kwak, S. G., & Kim, J. H. (2017). Central limit theorem: The cornerstone of modern statistics. *Korean Journal of Anesthesiology*, 70(2), 144–156. <https://doi.org/10.4097/kjae.2017.70.2.144>
- Lee, P. M., & Wahal, S. (2004). Grandstanding, certification and the underpricing of venture capital backed IPOs. *Journal of Financial Economics*, 73(2), 375–407. <https://doi.org/10.1016/j.jfineco.2003.09.003>
- Linton, O., & Shankar, S. (2023). Financial Econometrics Models and Methods, by Oliver Linton (Cambridge University Press, Cambridge, 2019), pp. xxvii +555. *Economic Record*, 99(325), 322–323. <https://doi.org/10.1111/1475-4932.12741>
- Liu, X., & Ritter, J. R. (2011). Local underwriter oligopolies and IPO underpricing [Place: Amsterdam Publisher: Elsevier B.V]. *Journal of financial economics*, 102(3), 579–601. <https://doi.org/10.1016/j.jfineco.2011.01.009>
- Loughran, T., & Ritter, J. (2004). Why Has IPO Underpricing Changed over Time? [Publisher: [Financial Management Association International, Wiley]]. *Financial Management*, 33(3), 5–37. Retrieved January 14, 2024, from <https://www.jstor.org/stable/3666262>
- Loughran, T., & Ritter, J. R. (2002). Why Don't Issuers Get Upset About Leaving Money on the Table in IPOs? *The Review of Financial Studies*, 15(2), 413–444. <https://doi.org/10.1093/rfs/15.2.413>

- Lowry, M., & Schwert, G. W. (2004). Is the IPO pricing process efficient? *Journal of Financial Economics*, 71(1), 3–26. [https://doi.org/10.1016/S0304-405X\(03\)00205-8](https://doi.org/10.1016/S0304-405X(03)00205-8)
- M, Tsagrisa, A, Alenazib, & K, Verrouc. (2020). Hypothesis testing for two population means: Parametric or non-parametric test? [Publisher: Taylor & Francis]. *Journal of statistical computation and simulation*. <https://doi.org/10.1080/00949655.2019.1677659>
- Megginson, W. L., & Weiss, K. A. (1991). Venture Capitalist Certification in Initial Public Offerings [_eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-6261.1991.tb03770.x>]. *The Journal of Finance*, 46(3), 879–903. <https://doi.org/10.1111/j.1540-6261.1991.tb03770.x>
- Metrick, A. (2006, January). *Venture Capital and the Finance of Innovation by Andrew Metrick* (27075th edition). Wiley,2006.
- PitchBook. (2023, December). The IPO process explained: A guide to going public | PitchBook [Section: PitchBook blog]. Retrieved June 24, 2024, from <https://pitchbook.com/blog/ipo-process-explained>
- Porta, R. L., Lopez-de-Silanes, F., Shleifer, A., & vishny robert w., r. w. (1998). Law and Finance [Publisher: The University of Chicago Press]. *Journal of Political Economy*, 106(6), 1113–1155. <https://doi.org/10.1086/250042>
- Ritter, J. R. (1991). The Long-Run Performance of initial Public Offerings [_eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-6261.1991.tb03743.x>]. *The Journal of Finance*, 46(1), 3–27. <https://doi.org/10.1111/j.1540-6261.1991.tb03743.x>
- Ritter, J. R., Gahng, M., & Zhang, D. (2023). SPACs. *The Review of Financial Studies*, 36(9), 3463–3501. <https://doi.org/10.1093/rfs/hhad019>
- Saunders, M., Lewis, P., & Thornhill, A. (2016). *Research methods for business students: 7. ed* (Seventh edition.). Pearson.

Appendices

A Comprehensive Results

Bloomberg Data

Field	Description
Announced Date	The date when the offer was announced.
Issuer Ticker	The stock ticker of the issuing company.
Issuer Name	The name of the issuing company.
Offer Size (M)	The size of the offer in millions.
Offer Type	The type of offer (e.g., IPO, SEO).
Country/Region	The country or region where the offer was made.
Offer To 1st Close	The performance from the offer date to the first close.
Offer To 1st 6 Months	The performance from the offer date to the first 6m.
Offer To 1st Year	The performance from the offer date to the first year.
Offer To Month 3	The performance from the offer date to the third month.
Industry Group	The industry group of the issuing company.
Industry Sector	The industry sector of the issuing company.
Action Type	The type of action taken (e.g., Equity Offering).

Table A.1: Description of Bloomberg Data Fields

PitchBook Data

Table	Description
CompanyID	The unique identifier for each company.
LastKnownValuationDealType	The type of deal for the company's last known valuation.
FirstFinancingDealType	The type of deal for the company's first financing.
Universe	The classification or category of the company within PitchBook's database (e.g., Venture Capital, IPO).
OwnershipStatus	The ownership status of the company (e.g., Privately Held, Public).
CompanyFinancingStatus	The current financing status of the company (e.g., Venture Capital-Backed).
InvestorType	The type of investors involved with the company.
InvestorSince	The date since the investors have been involved with the company.
CompanyInvestorRelation	The nature of the relationship between the company and its investors.

Table A.2: Description of data fields exported from PitchBook

Hypotheses 3

Test	Group	p-value
Mann-Whitney U	VC-backed companies	0.0337
	Non-VC-backed companies	0.1387
T-test	VC-backed companies	0.0134
	Non-VC-backed companies	0.2145

Table A.3: P-values of Mann-Whitney U and T-tests for VC-backed and Non-VC-backed companies (3 years before and after 2011).

Difference-in-Differences Analysis

Variable	Coefficient	p-value
const	12.7956	0.000
PostLaw	-2.7499	0.198
VCBacked	7.3244	0.102
PostLaw*VCBacked	-10.2003	0.060

Table A.4: Hypotheses 3 results of Difference-in-Differences analysis for 3 years before and after 2011.

Hypotheses 4

Variable	Common Law	French Civil Law	German Civil Law	Scandinavian Civil Law
const	5.9923***	5.4702***	19.4820***	3.4314
PostLaw	2.9838	-2.2712	-0.6578	3.6020
VCBacked	17.6231***	6.7176	-1.6698	31.1303**
PostLaw*VCBacked	-19.8749***	0.6474	-14.1294	-37.3065
R-squared	0.022	0.016	0.000	0.029
Adj. R-squared	0.011	0.006	-0.006	0.004
F-statistic	3.067**	2.173*	0.8223	1.382
Prob (F-statistic)	0.0285**	0.0913*	0.482	0.252

Notes:

*** p<0.01, ** p<0.05, * p<0.1.

Table A.5: Hypotheses 4 results of Difference-in-Differences analysis for 3 years before and after 2011 with legal regions.

Legal Origin

Country	SHR	EW	CRR	EW	AS	EW	EQ	EW	Avg	Legal Origin
Turkey	2	0.40	2	0.50	50	0.60	6	0.60	0.53	French Civil Law
Greece	2	0.40	2	0.50	50	0.60	6.5	0.65	0.54	French Civil Law
Portugal	3	0.60	1	0.25	52	0.63	6.75	0.68	0.54	French Civil Law
Spain	2	0.40	2	0.50	54	0.65	7	0.70	0.56	French Civil Law
Italy	1	0.20	2	0.50	62	0.75	7.5	0.75	0.55	French Civil Law
France	3	0.60	0	0	64	0.77	8.75	0.88	0.56	French Civil Law
Ireland	4	0.80	1	0.25	72	0.87	9.25	0.93	0.71	Common Law
Finland	3	0.60	1	0.25	70	0.84	9.25	0.93	0.66	Scandinavian Law
Norway	4	0.80	2	0.50	68	0.82	9.25	0.93	0.76	Scandinavian Law
Sweden	3	0.60	3	0.75	70	0.84	9.25	0.93	0.78	Scandinavian Law
Belgium	0	0	2	0.50	61	0.73	9.75	0.98	0.55	French Civil Law
United Kingdom	5	1.00	4	1.00	78	0.94	10	1.00	0.99	Common Law
Netherlands	2	0.40	2	0.50	70	0.84	10	1.00	0.69	French Civil Law
Austria	2	0.40	3	0.75	62	0.75	10	1.00	0.72	German Civil Law
Germany	1	0.20	3	0.75	62	0.75	10	1.00	0.67	German Civil Law
Switzerland	3	0.60	1	0.25	72	0.87	10	1.00	0.68	German Civil Law
Denmark	2	0.40	3	0.75	62	0.75	10	1.00	0.72	Scandinavian Law

Table A.6: Scores for each legal region equal weighted

Explanation of Scores:

- **SHR:** Shareholder Rights Score
- **CRR:** Creditor Rights Score
- **AS:** Accounting Standards Score
- **EQ:** Enforcement Quality Score
- **Avg:** Average Score
- **EW:** EW Score (Equal Weighted Score)

Legal Origins:

- **Common Law:** Typically offers the strongest investor protections and enforcement mechanisms.
- **French Civil Law:** Generally provides weaker protections and enforcement.
- **German Civil Law:** Intermediate level of protection and enforcement.
- **Scandinavian Law:** Also provides intermediate levels of protection and enforcement.